

SALMIYA, Kuwait

WMO: 405850

Lat: 29.350N

Lon: 48.100E

Elev: 8

StdP: 101.23

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.0	10.5	-5.2	2.4	13.8	-2.7	3.0	15.3	19.3	16.4	15.5	13.6	6.3	310	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	43.1	22.0	41.8	22.5	40.7	22.4	31.8	34.5	30.9	34.4	30.2	34.1	5.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.2	29.2	34.2	30.1	27.3	33.9	29.1	25.9	33.8	108.7	34.6	103.3	34.3	99.7	34.1	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	13.9	12.1	DB	6.6	46.0	1.9	1.0	5.2	46.7	4.1	47.3	3.0	47.9	1.6	48.6	(4)
(5)			WB	3.0	32.4	1.9	1.5	1.6	33.5	0.5	34.4	-0.5	35.3	-1.9	36.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	14.8	16.4	20.4	24.8	31.2	35.9	37.1	36.9	34.7	29.9	22.2	17.2	(6)
(7)		DBStd	8.45	2.48	2.28	2.42	2.33	2.57	2.23	1.94	1.86	1.91	2.35	2.96	2.22	(7)
(8)		HDD10.0	2	2	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	229	112	58	7	0	0	0	0	0	0	0	4	48	(9)
(10)		CDD10.0	6152	150	181	322	443	657	777	840	834	741	618	367	223	(10)
(11)		CDD18.3	3337	1	6	71	193	399	527	582	576	491	359	121	12	(11)
(12)		CDH23.3	49181	0	1	156	1231	5655	8875	9979	9811	7982	4896	593	2	(12)
(13)		CDH26.7	32962	0	0	28	240	3217	6475	7499	7331	5583	2511	78	0	(13)
(14)	Wind	WSAvg	5.1	5.1	4.9	5.1	4.7	5.0	6.3	5.8	5.5	5.2	4.9	4.7	4.5	(14)
(15)	Precipitation	PrecAvg	182	35	23	31	19	5	1	1	0	4	32	30	(15)	
(16)		PrecMax	298	103	81	97	52	20	2	3	2	1	26	182	90	(16)
(17)		PrecMin	82	1	1	1	1	0	0	0	0	0	0	1	1	(17)
(18)		PrecStd	61	26	20	24	15	5	1	1	0	0	6	39	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	23.1	29.6	32.5	39.8	44.8	44.7	43.8	42.1	38.1	29.9	23.1	(19)
(20)			MCWB	13.7	13.5	15.2	17.6	20.5	20.6	21.8	22.5	21.9	21.5	17.7	15.3	(20)
(21)		2%	DB	19.2	21.5	26.6	30.4	37.7	42.7	42.9	42.5	40.0	35.7	28.3	21.7	(21)
(22)			MCWB	14.3	13.5	15.7	18.1	20.9	21.3	22.3	22.7	22.3	20.6	20.8	16.1	(22)
(23)		5%	DB	18.6	20.4	24.8	29.1	36.3	40.9	41.8	41.4	38.8	34.3	27.5	20.9	(23)
(24)			MCWB	14.5	13.6	15.9	18.5	20.8	21.4	22.6	23.1	22.6	21.1	20.6	16.6	(24)
(25)		10%	DB	17.9	19.3	23.5	27.9	35.0	39.5	40.7	40.2	37.8	33.2	26.6	20.3	(25)
(26)			MCWB	14.2	14.1	16.0	18.6	21.0	21.4	22.5	23.3	22.8	22.1	18.8	16.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	18.4	21.3	23.5	27.2	29.4	32.6	34.1	32.4	29.0	25.7	20.1	(27)
(28)			MCD B	18.5	18.8	24.0	27.3	31.4	33.3	33.8	34.5	34.4	32.0	27.6	21.2	(28)
(29)		2%	WB	16.7	17.4	19.7	22.5	26.4	28.3	31.3	32.3	30.7	28.0	23.8	19.0	(29)
(30)			MCD B	18.1	18.5	22.4	26.1	30.6	33.5	34.4	34.8	34.1	31.5	27.4	20.7	(30)
(31)		5%	WB	15.9	16.5	19.2	21.8	25.6	26.9	30.1	31.3	29.9	27.2	22.0	18.0	(31)
(32)			MCD B	17.5	18.3	21.7	25.8	30.0	33.4	33.7	34.5	33.9	30.9	25.8	20.1	(32)
(33)		10%	WB	15.2	15.8	18.6	21.2	24.8	25.6	29.3	30.7	28.8	26.1	20.5	16.9	(33)
(34)			MCD B	17.1	18.0	21.3	25.4	29.9	34.5	33.6	34.5	33.6	30.2	24.6	19.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.9	4.1	4.4	4.3	5.3	6.8	6.7	5.7	5.4	4.4	4.4	4.2	(35)
(36)			MCDBR	4.2	5.3	6.7	6.4	7.1	8.8	8.3	7.8	7.1	6.4	5.2	4.5	(36)
(37)			MCWBR	3.4	3.3	3.7	4.6	4.9	4.8	5.1	5.2	5.3	5.3	4.7	3.9	(37)
(38)		5% WB	MCD BR	3.0	3.1	3.4	4.0	4.4	5.9	5.5	4.2	3.5	3.3	3.7	2.9	(38)
(39)			MCWBR	4.0	3.4	3.8	4.7	5.7	7.4	7.2	7.5	7.1	6.8	6.0	4.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.511	0.557	0.612	0.699	0.699	0.653	0.678	0.628	0.572	0.576	0.548	0.512	(40)	
(41)		taud	1.844	1.752	1.635	1.531	1.535	1.585	1.572	1.662	1.759	1.756	1.801	1.852	(41)	
(42)		Ebn at Noon	691	701	696	655	659	687	668	697	718	675	650	664	(42)	
(43)		Edh at Noon	175	209	250	286	286	270	273	247	217	205	182	167	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.40	4.25	5.37	5.86	6.76	7.52	7.21	6.86	6.10	4.85	3.54	3.13	(44)	
(45)		RadStd	0.24	0.28	0.23	0.31	0.31	0.25	0.19	0.16	0.21	0.19	0.28	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+0.33	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	+94	+107

Nomenclature: See separate page